UNIVERSITI MALAYA

NFMN

SIM 1005 / SIN 1002

Fundamentals of Spreadsheets / Introduction to Worksheet

Lecture 6: System of Differential Equations

By

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Objectives

- To define basic types of differential equations
- To apply Euler method for solving ordinary differential equations subject to initial conditions.
- To produce Euler solutions of the ordinary differential equation system by using Microsoft Excel.

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Differential Equation

- There are 2 types of differential equation:
  1) Ordinary differential equation (ODE)
  2) Partial differential equation (PDE)*

- An ODE has only 1 independent variable i.e.

$$\frac{dy}{dx} = f(x, y)$$

- A PDE has >1 independent variables i.e.

$$\frac{\partial^2 t}{\partial x \partial y} = f(x, y, t), \frac{\partial^3 t}{\partial x \partial y \partial z} = f(x, y, z, t)$$

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Euler Method

- A numerical technique to solve 1st order ODE of the form:

$$\frac{dy}{dx} = f(x, y), y(0) = y_0.$$

- Formula of Euler method:

$$y_{i+1} = y_i + f(x_i, y_i)\Delta x,$$

for $i = 0, .., n$ where

$$\Delta x = (x_{i+1} - x_i).$$

Original notes by NFM Noor

Example 1

Consider

$$\frac{dy}{dx} + 2y = 1.3e^{-x}, y(0) = 5.$$

Using Euler method, find the solution of $y(2)$ up to 4 decimal places if $\Delta x = 0.2$.

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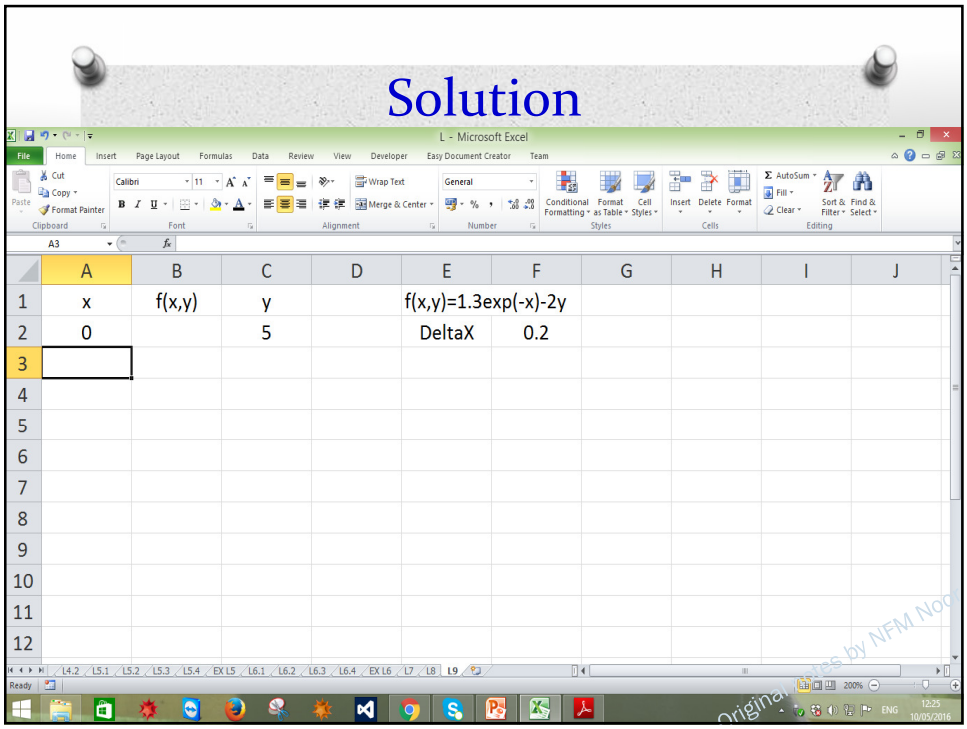
Solution

$$\frac{dy}{dx} + 2y = 1.3e^{-x}, y(0) = 5$$

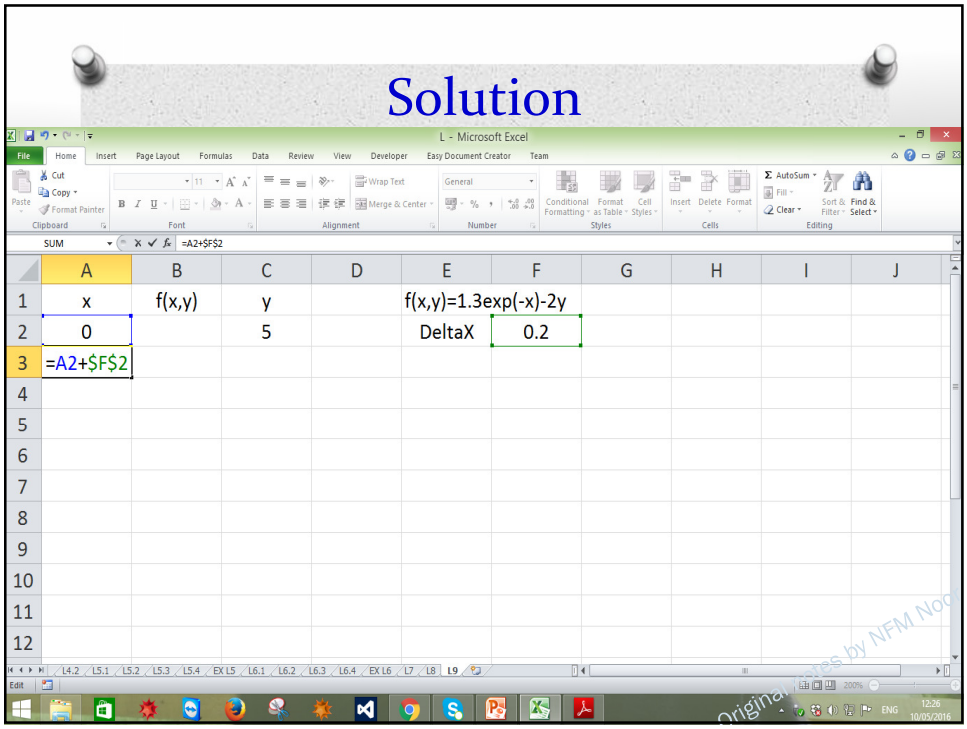
$$\frac{dy}{dx} = 1.3e^{-x} - 2y$$
$$f(x, y) = 1.3e^{-x} - 2y$$
$$x_0 = 0, y_0 = 5$$
$$\Delta x = 0.2$$

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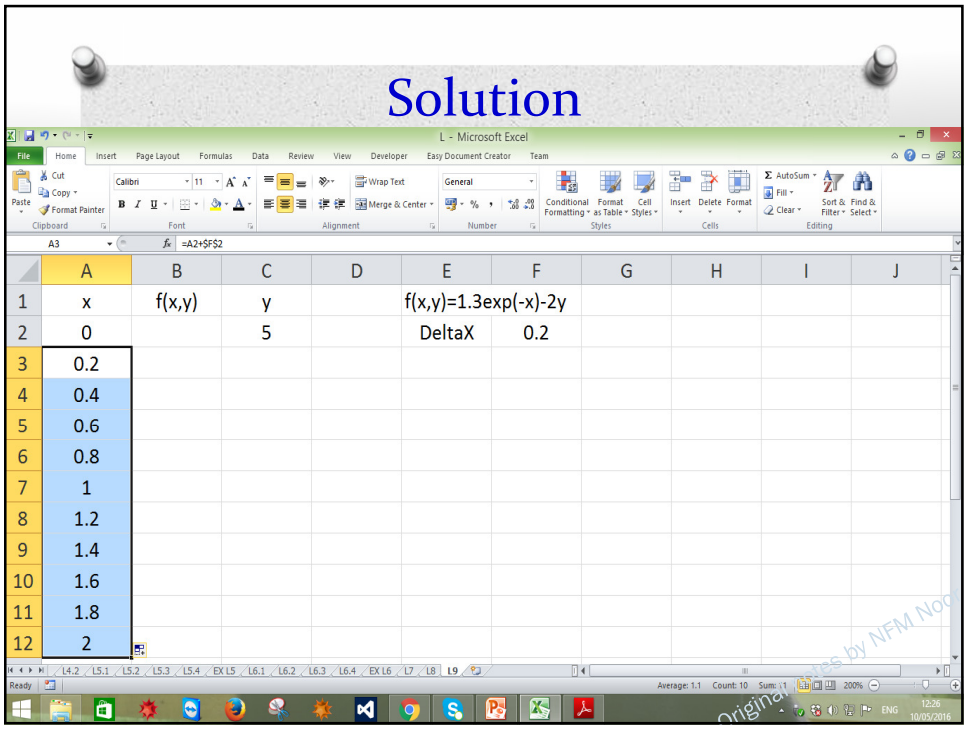
Solution



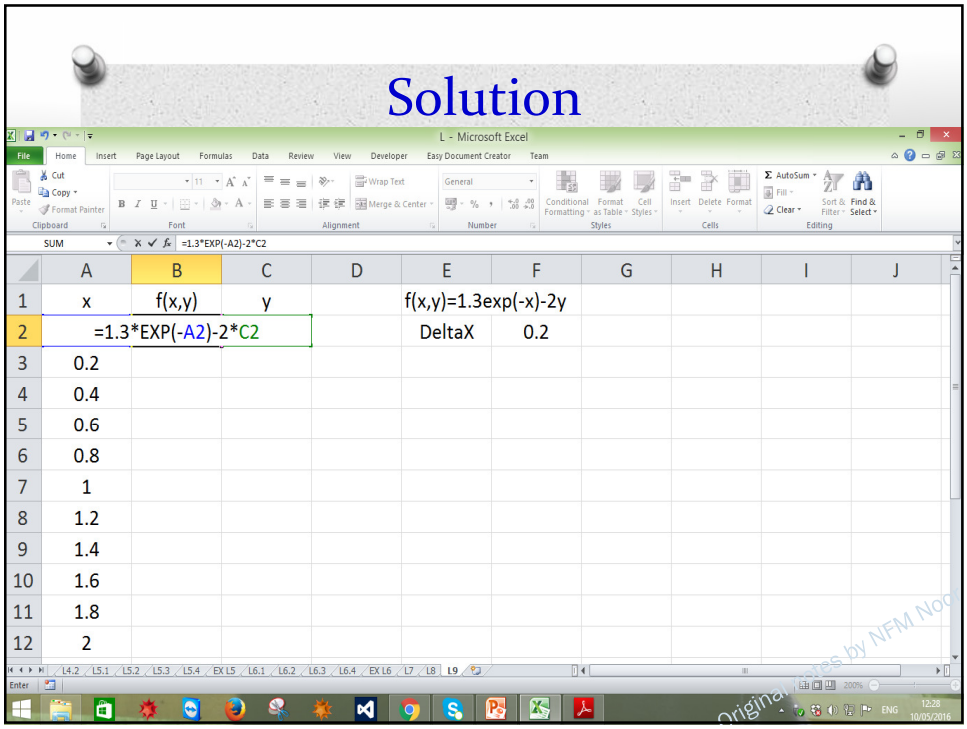
Solution



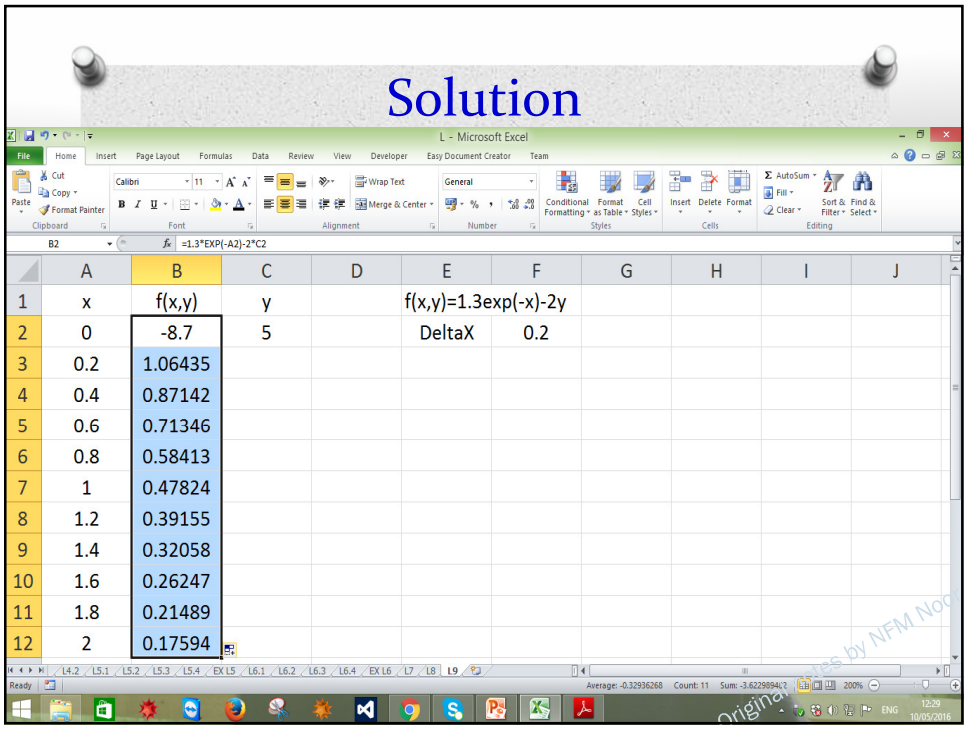
Solution



Solution

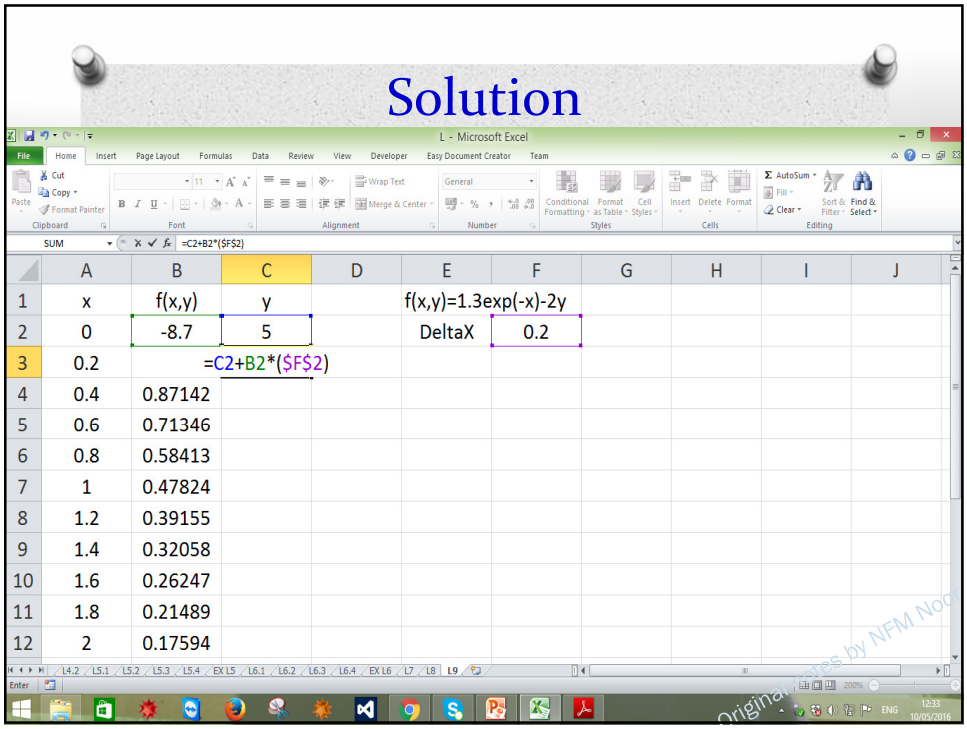


Solution



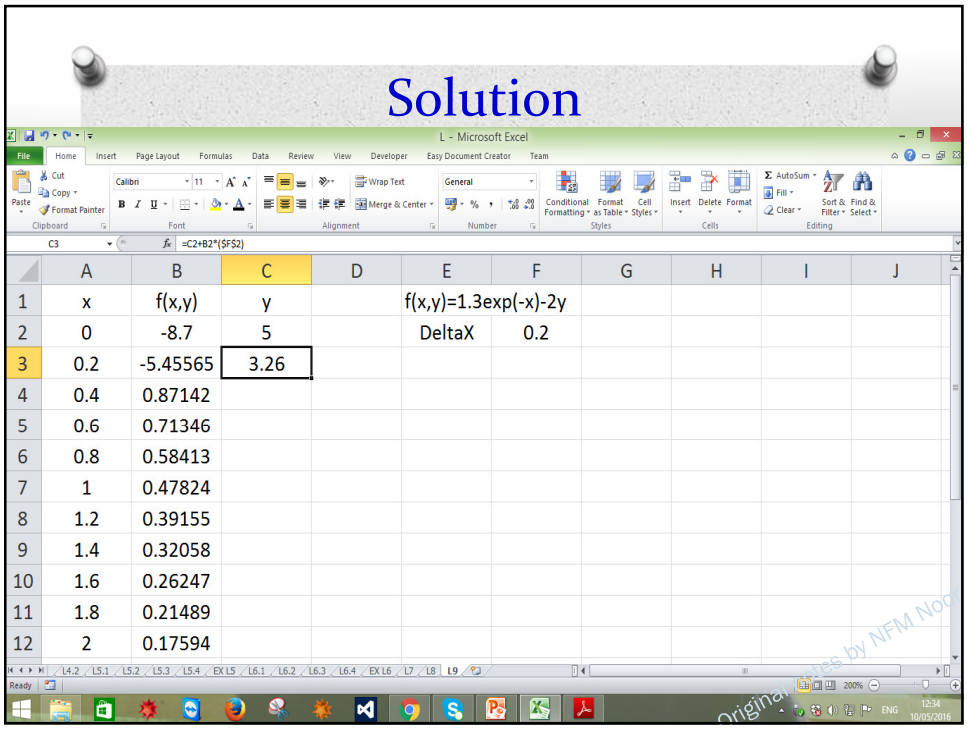
	A	B	C	D	E	F	G	H	I	J
1	x	f(x,y)	y		f(x,y)=1.3exp(-x)-2y					
2	0	-8.7	5		DeltaX	0.2				
3	0.2	1.06435								
4	0.4	0.87142								
5	0.6	0.71346								
6	0.8	0.58413								
7	1	0.47824								
8	1.2	0.39155								
9	1.4	0.32058								
10	1.6	0.26247								
11	1.8	0.21489								
12	2	0.17594								

Solution



	A	B	C	D	E	F	G	H	I	J
1	x	f(x,y)	y		f(x,y)=1.3exp(-x)-2y					
2	0	-8.7	5		DeltaX	0.2				
3	0.2	=C2+B2*(F2)								
4	0.4	0.87142								
5	0.6	0.71346								
6	0.8	0.58413								
7	1	0.47824								
8	1.2	0.39155								
9	1.4	0.32058								
10	1.6	0.26247								
11	1.8	0.21489								
12	2	0.17594								

Solution

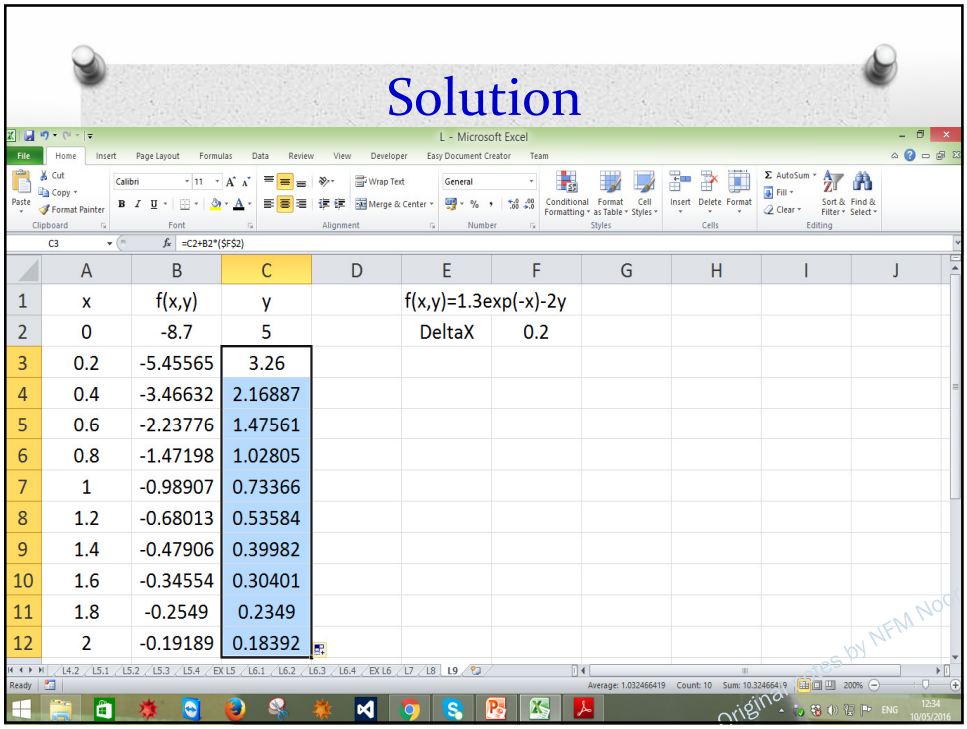


L - Microsoft Excel

=C2+B2*(F2)

	A	B	C	D	E	F	G	H	I	J
1	x	f(x,y)	y		f(x,y)=1.3exp(-x)-2y					
2	0	-8.7	5		DeltaX	0.2				
3	0.2	-5.45565	3.26							
4	0.4	0.87142								
5	0.6	0.71346								
6	0.8	0.58413								
7	1	0.47824								
8	1.2	0.39155								
9	1.4	0.32058								
10	1.6	0.26247								
11	1.8	0.21489								
12	2	0.17594								

Solution

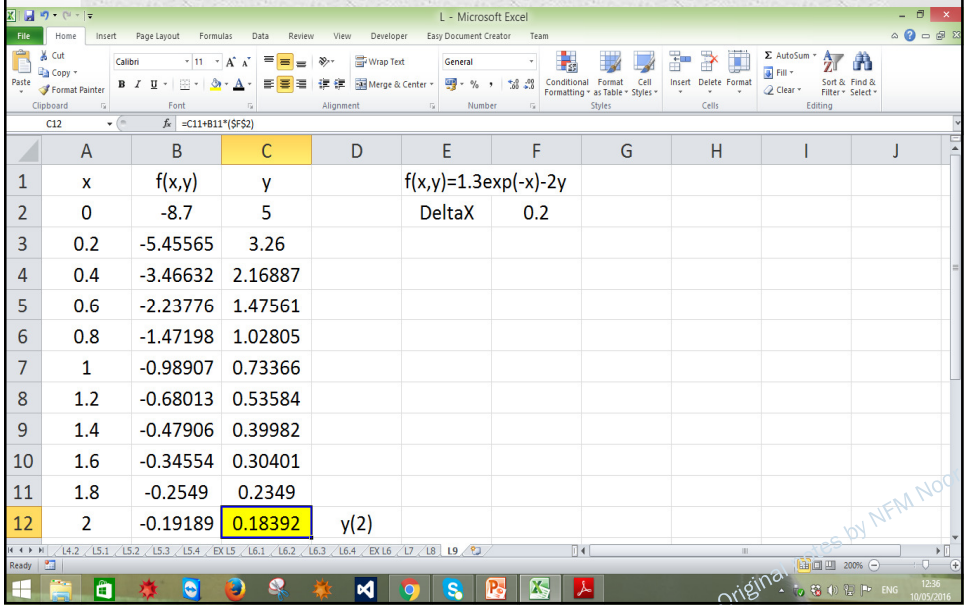


L - Microsoft Excel

=C2+B2*(F2)

	A	B	C	D	E	F	G	H	I	J
1	x	f(x,y)	y		f(x,y)=1.3exp(-x)-2y					
2	0	-8.7	5		DeltaX	0.2				
3	0.2	-5.45565	3.26							
4	0.4	-3.46632	2.16887							
5	0.6	-2.23776	1.47561							
6	0.8	-1.47198	1.02805							
7	1	-0.98907	0.73366							
8	1.2	-0.68013	0.53584							
9	1.4	-0.47906	0.39982							
10	1.6	-0.34554	0.30401							
11	1.8	-0.2549	0.2349							
12	2	-0.19189	0.18392							

Solution



L - Microsoft Excel

=C11+B11*(F2)

	A	B	C	D	E	F	G	H	I	J
1	x	f(x,y)	y		f(x,y)=1.3exp(-x)-2y					
2	0	-8.7	5		DeltaX	0.2				
3	0.2	-5.45565	3.26							
4	0.4	-3.46632	2.16887							
5	0.6	-2.23776	1.47561							
6	0.8	-1.47198	1.02805							
7	1	-0.98907	0.73366							
8	1.2	-0.68013	0.53584							
9	1.4	-0.47906	0.39982							
10	1.6	-0.34554	0.30401							
11	1.8	-0.2549	0.2349							
12	2	-0.19189	0.18392	y(2)						

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Exercise 1

Consider

$$\frac{dy}{dx} - 5y = sin(x) + 0.7, y(1) = -4.$$

Using Euler method, find the solution of $y(5)$ up to 4 decimal places if $\Delta x = 0.01$.

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Solution

$$\frac{dy}{dx} - 5y = sin(x) + 0.7, y(1) = -4.$$

$$\frac{dy}{dx} = sin(x) + 0.7 + 5y$$
$$f(x, y) = sin(x) + 0.7 + 5y$$
$$x_0 = 1, \quad y_0 = -4$$
$$\Delta x = 0.01$$

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Example 2

Consider

$$e^y \frac{dy}{dx} + xy = cosh(4x), y(0.2) = 2.$$

Using Euler method, find the solution of $y(0.7)$ up to 4 decimal places if $\Delta x = 0.05$.

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Solution

$$e^y \frac{dy}{dx} + xy = cosh(4x), y(0.2) = 2.$$

$$\frac{dy}{dx} = \frac{cosh(4x) - xy}{e^y}$$

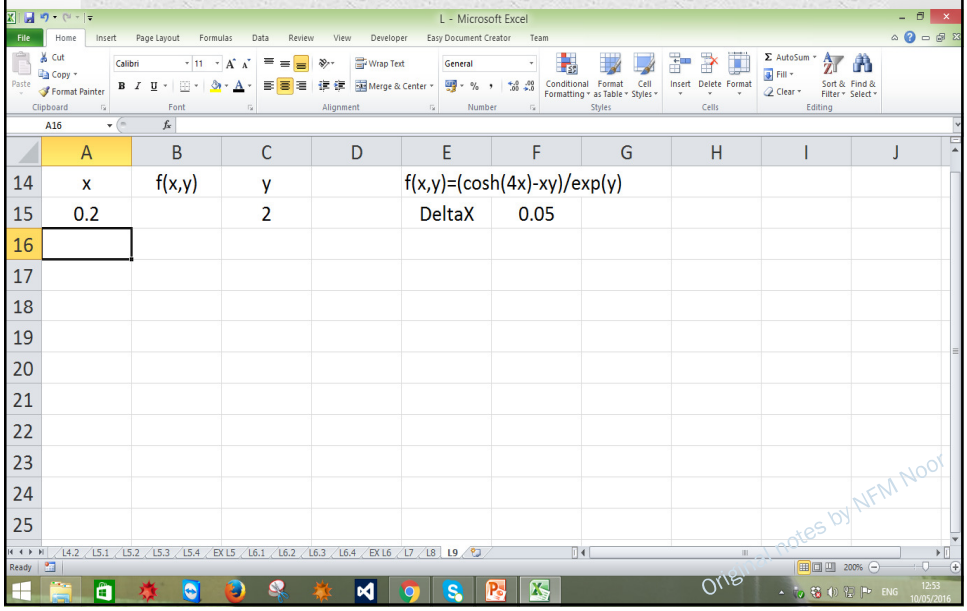
$$f(x,y) = \frac{cosh(4x) - xy}{e^y}$$

$$x_0 = 0.2, y_0 = 2$$

$$\Delta x = 0.05$$

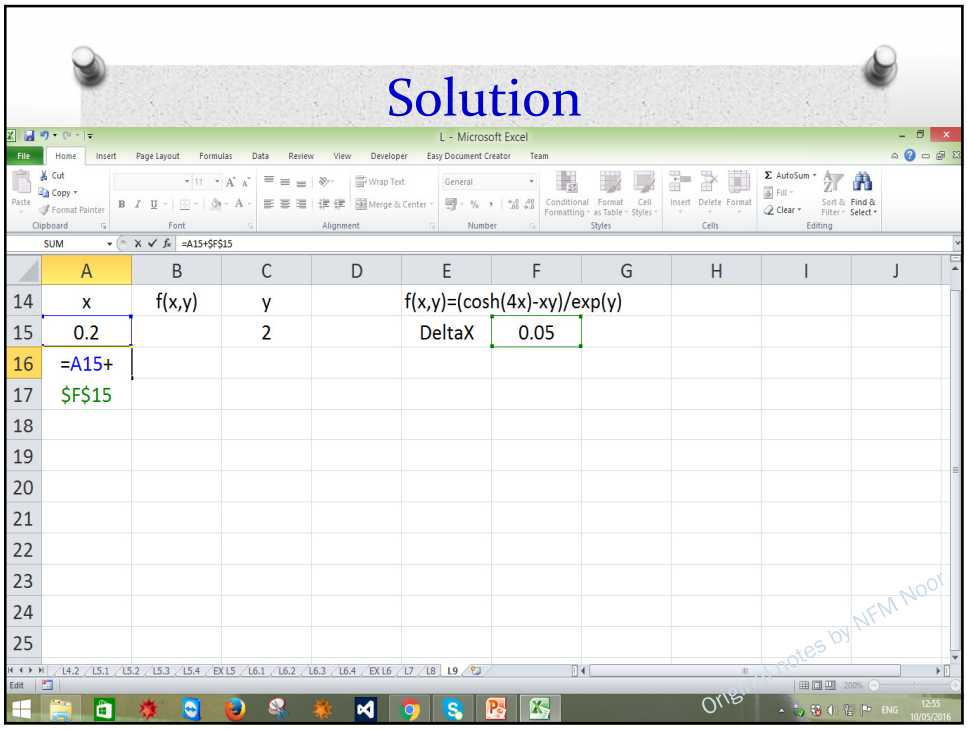
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Solution

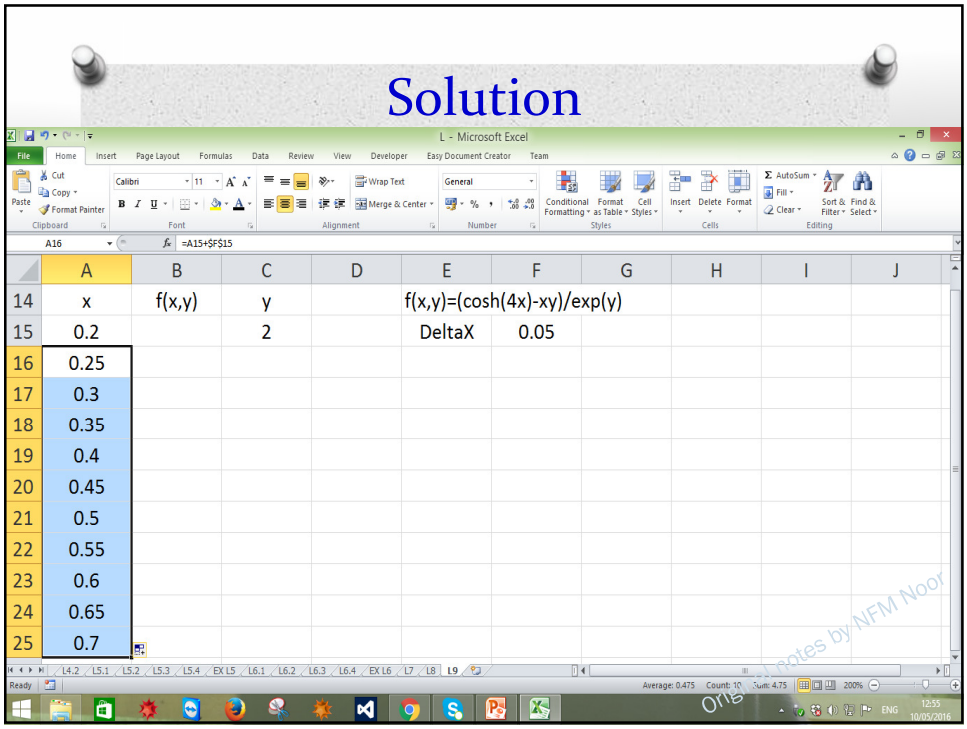


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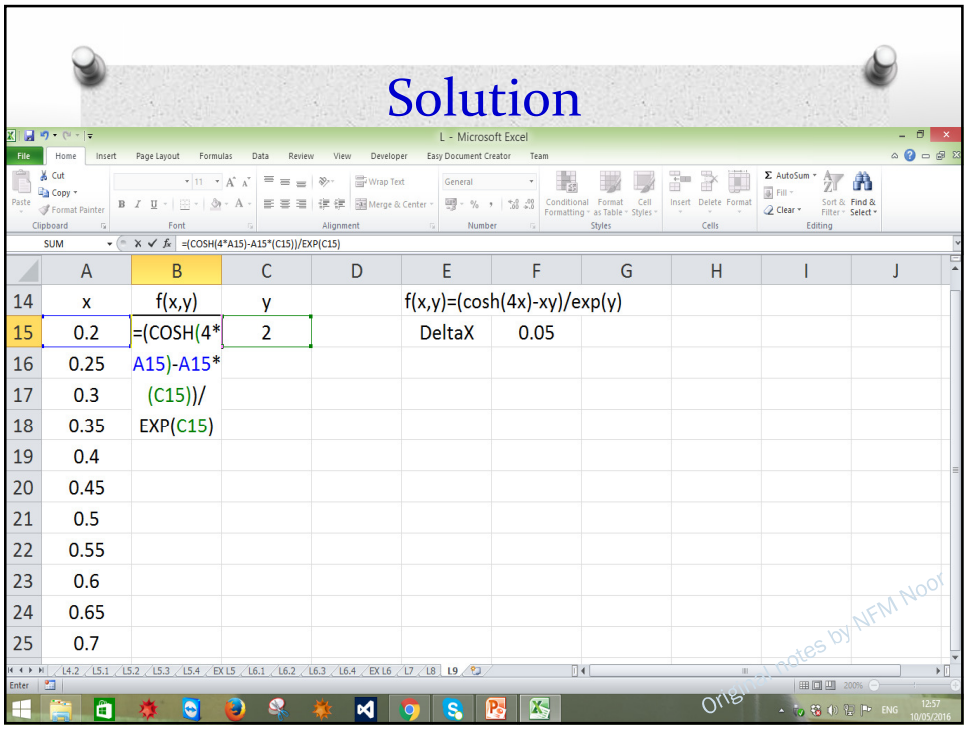
Solution



Solution

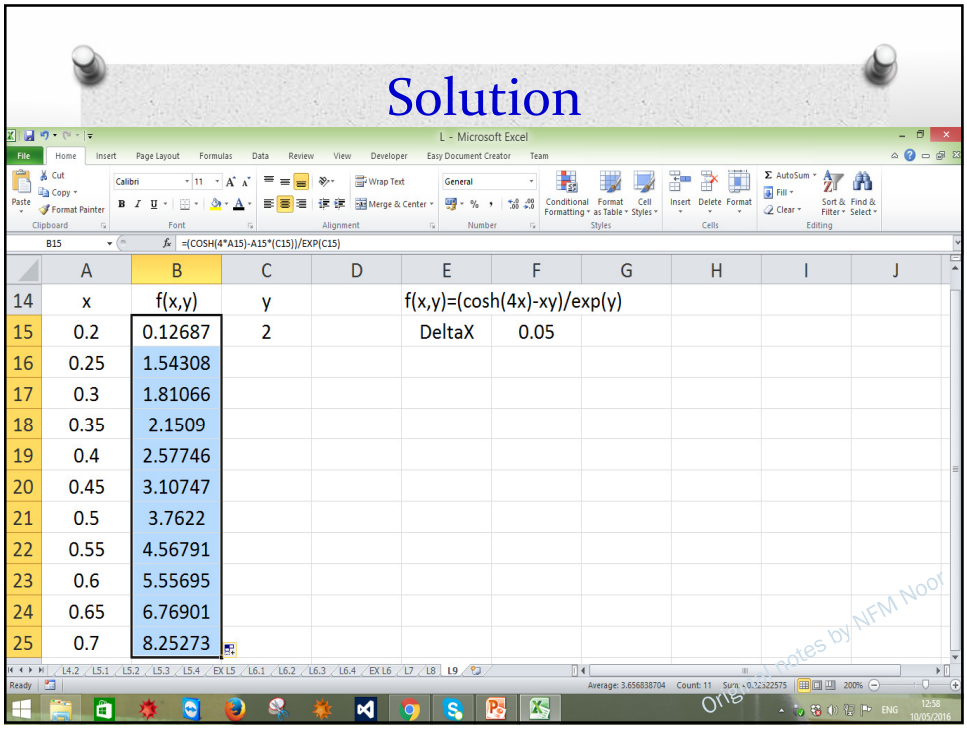


Solution



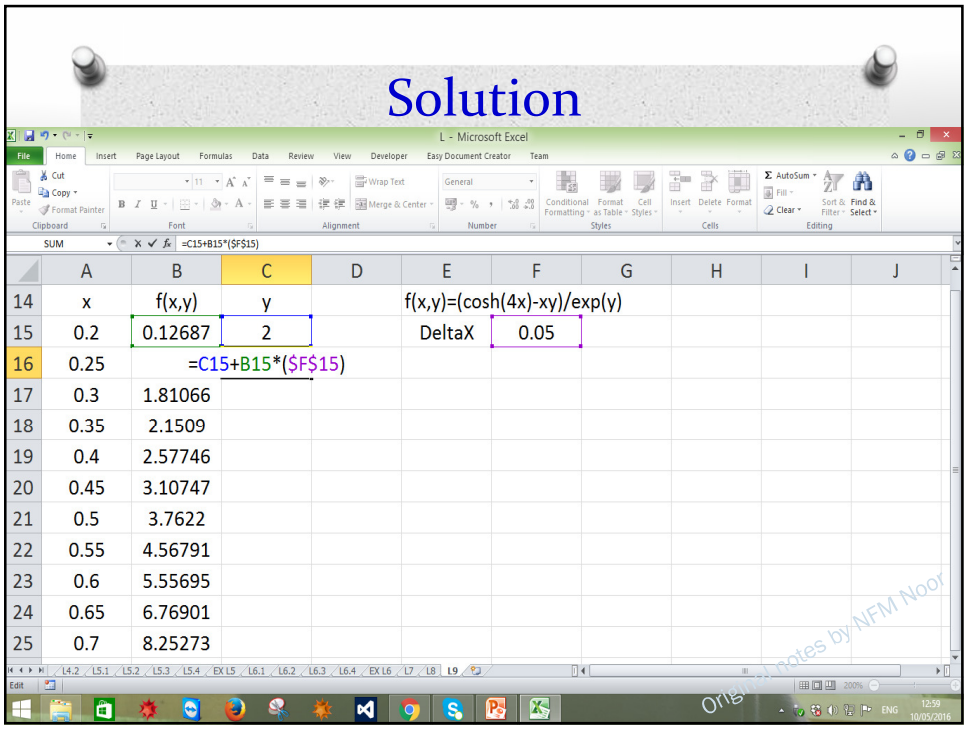
	A	B	C	D	E	F	G	H	I	J
14	x	f(x,y)	y		f(x,y)=(cosh(4x)-xy)/exp(y)					
15	0.2	=(COSH(4*	2		DeltaX	0.05				
16	0.25	A15)-A15*								
17	0.3	(C15))/								
18	0.35	EXP(C15)								
19	0.4									
20	0.45									
21	0.5									
22	0.55									
23	0.6									
24	0.65									
25	0.7									

Solution



	A	B	C	D	E	F	G	H	I	J
14	x	f(x,y)	y		f(x,y)=(cosh(4x)-xy)/exp(y)					
15	0.2	0.12687	2		DeltaX	0.05				
16	0.25	1.54308								
17	0.3	1.81066								
18	0.35	2.1509								
19	0.4	2.57746								
20	0.45	3.10747								
21	0.5	3.7622								
22	0.55	4.56791								
23	0.6	5.55695								
24	0.65	6.76901								
25	0.7	8.25273								

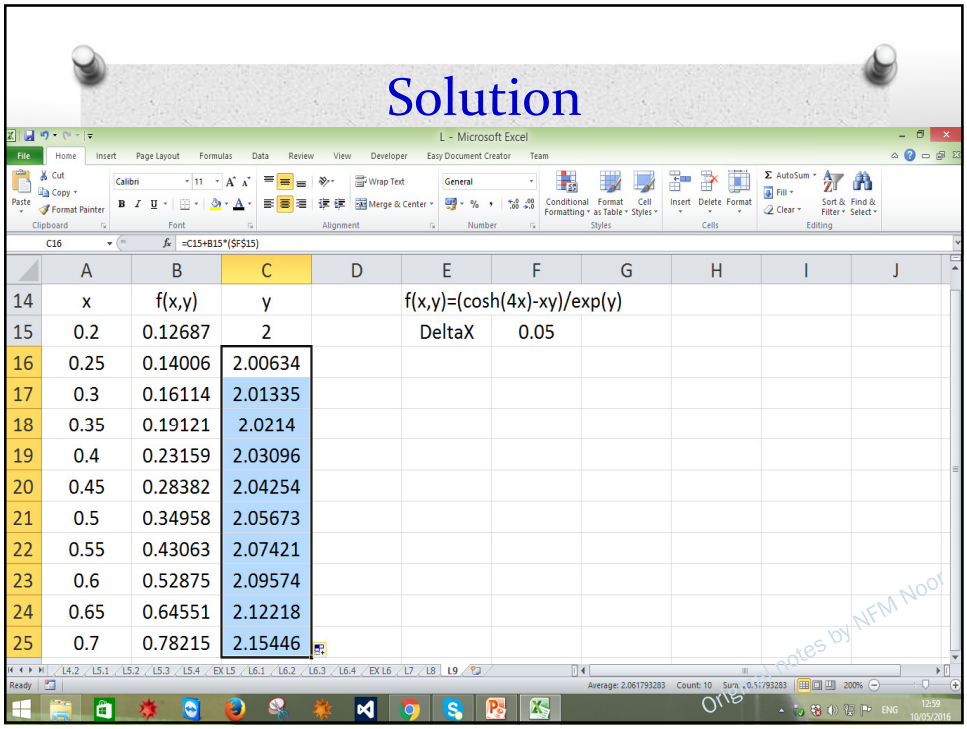
Solution



	A	B	C	D	E	F	G	H	I	J
14	x	f(x,y)	y		f(x,y)=(cosh(4x)-xy)/exp(y)					
15	0.2	0.12687	2		DeltaX	0.05				
16	0.25	=C15+B15*(F15)								
17	0.3	1.81066								
18	0.35	2.1509								
19	0.4	2.57746								
20	0.45	3.10747								
21	0.5	3.7622								
22	0.55	4.56791								
23	0.6	5.55695								
24	0.65	6.76901								
25	0.7	8.25273								

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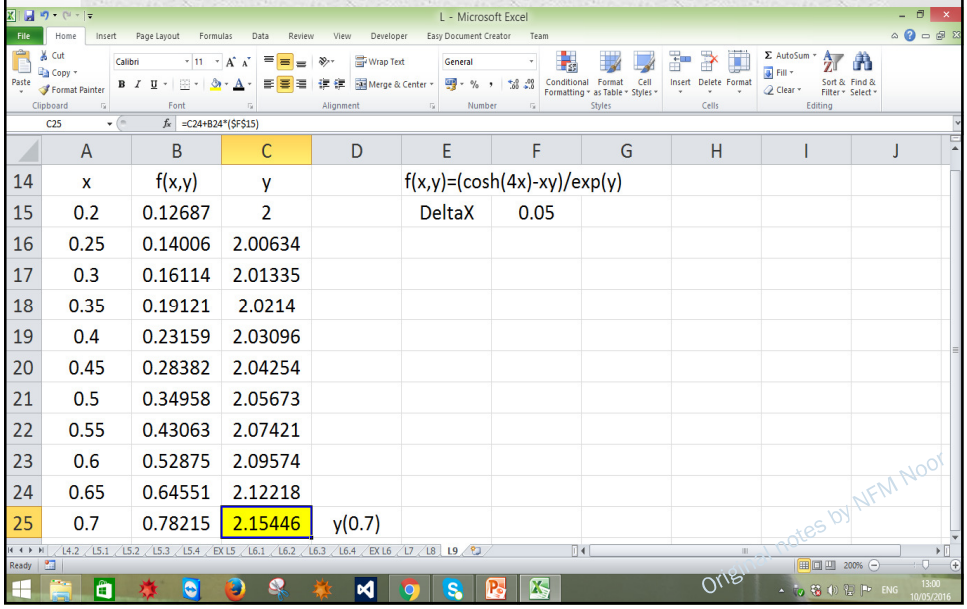
Solution



	A	B	C	D	E	F	G	H	I	J
14	x	f(x,y)	y		f(x,y)=(cosh(4x)-xy)/exp(y)					
15	0.2	0.12687	2		DeltaX	0.05				
16	0.25	0.14006	2.00634							
17	0.3	0.16114	2.01335							
18	0.35	0.19121	2.0214							
19	0.4	0.23159	2.03096							
20	0.45	0.28382	2.04254							
21	0.5	0.34958	2.05673							
22	0.55	0.43063	2.07421							
23	0.6	0.52875	2.09574							
24	0.65	0.64551	2.12218							
25	0.7	0.78215	2.15446							

Original notes by NFM Noor

Solution



	A	B	C	D	E	F	G	H	I	J
14	x	f(x,y)	y		f(x,y)=(cosh(4x)-xy)/exp(y)					
15	0.2	0.12687	2		DeltaX	0.05				
16	0.25	0.14006	2.00634							
17	0.3	0.16114	2.01335							
18	0.35	0.19121	2.0214							
19	0.4	0.23159	2.03096							
20	0.45	0.28382	2.04254							
21	0.5	0.34958	2.05673							
22	0.55	0.43063	2.07421							
23	0.6	0.52875	2.09574							
24	0.65	0.64551	2.12218							
25	0.7	0.78215	2.15446	y(0.7)						

Original notes by NFM Noor

Exercise 2

Consider

$$2e^{y}\frac{dy}{dx} + 3x^2y^2 = -5sin(2x),$$
$$y(0) = -1.2.$$

Using Euler method, find the solution of $y(0.312)$ up to 4 decimal places if $\Delta x = 0.002$.

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Solution

$$2e^{y}\frac{dy}{dx} + 3x^2y^2 = -5sin(2x), y(0) = -1.2.$$

$$\frac{dy}{dx} = \frac{-3x^2y^2 - 5sin(2x)}{2e^{y}}$$

$$f(x,y) = \frac{-3x^2y^2 - 5sin(2x)}{2e^{y}}$$

$$x_0 = 0, y_0 = -1.2$$

$$\Delta x = 0.002$$

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Summary

- Euler is a numerical iterative method for solving 1st order system of ordinary differential equations.
- Microsoft Excel can be used to generate Euler solutions for the 1st order system of ordinary differential equations easily.

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